Jay Doshi

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Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Relevant Coursework: Statistics, Kalman Filtering, Database Systems, Operating Systems, Software Engineering

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Collaborated with OT and process-control engineers to validate system integrations and support production cutovers.
- Diagnosed performance degradation via root-cause analysis of SolarWinds NPM metrics, resolving issues before production outages.
- Integrated server, VMware virtualized, and backup systems into stable hosting, keeping plant applications highly available.

Process Controls Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Engineered a threshold-based anomaly detector comparing live gauge measurements to plate specs, cutting defective escapes 30%.
- Shipped VB.NET journaling microservices to a clustered QMS app, restoring PLC-to-SQL Server TCP/IP data flow.
- Automated ETL pipelines moving live sensor data from legacy OpenVMS Alpha into SQL Server, cutting manual effort by 40%.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Managed 200+ IT assets through inventory tracking, device deployment, secure decommissioning, and life-cycle documentation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Reduced target navigation error by 40% by tuning a custom PID flight controller in a ROS2-Gazebo digital twin.
- Prototyped PID control designs in MATLAB, trialing gain settings for the PX4 drone flight controller.
- Improved control accuracy by 60% by analyzing flight-test data from 15+ hardware trials in Python to isolate failure patterns.
- Validated PX4 control logic in a ROS2-Gazebo simulation linking flight controller and simulator over MAVLink/UDP.

Professional Projects

Smart Obstacle Avoidance Vehicle | Embedded Systems, C++, Arduino UNO, TinkerCAD **Github Link**

- Developed embedded C++ software with ultrasonic sensors and servo motors for real-time obstacle detection and navigation.
- Implemented speed and distance tracking via IR sensor and rotating grid plate, achieving 85% measurement accuracy.
- Designed and tested a modular L293D motor-driver architecture, ensuring reliable integration and smooth motor control.

Multi-Agent Robotics Simulation | PX4-Autopilot, ROS 2 Humble, Gazebo Harmonic, Rviz2 **Project Link**

- Engineered a high-fidelity simulation pipeline using Gazebo Harmonic to test robotic controller logic in complex environments.
- Optimized collision physics and dynamics, ensuring 95%+ parity between simulation and physical hardware.
- Built multi-agent communication protocols, enabling synchronized control and real-time data feedback.

Automatic Street Light (LDR-Based) | Circuit Design, LDR, Soldering, Embedded Electronics **Course Project**

- Designed an LDR light-sensing circuit with a 2N2222A transistor and 1kΩ resistor to switch an LED on ambient light.
- Built the circuit by soldering all components onto a physical board, eliminating manual light control entirely.
- Achieved fast-response, low-power operation under a 5V supply using transistor base-voltage switching logic.

Technical Skills

Programming Languages: C++, Python, C, MATLAB, Java, SQL, Bash Scripting

Controls & Robotics: PID Control, PLCs, PX4, ROS2, Gazebo, MAVLink, QGroundControl, Rviz2, Simulation Testing

Electronics & Embedded: Embedded Systems, Arduino, Circuit Design, Soldering, KiCad, Sensors, Motor Drivers, TinkerCAD

Data & Analysis: Pandas, NumPy, Matplotlib, Root-Cause Analysis, SolarWinds NPM, SQL Server

Developer Tools: Git, Linux Command Line, VS Code, Jupyter Notebook, Docker